Phongsakon Mark Konrad

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Summary

Final-year BSc Software Engineering student at the University of Southern Denmark and incoming MPhil student in Machine Learning and Machine Intelligence at the University of Cambridge (October 2026). Eighteen papers across mechanistic interpretability, AI safety evaluation, software architecture, medical imaging, remote sensing, and data visualisation form the empirical foundation. The core questions I pursue are: how do models learn internal structure, what do those structures causally do, and how can we build models that are genuinely aligned rather than merely large? Current work includes circuit-level audits of fine-tuning-installed routing behaviour (NeurIPS 2026), evidential standards for safety claims, causal tracing of moral reasoning, and a cross-family study of chain-of-thought safety. I lead the full research lifecycle independently: problem formulation, experimental design, pipeline engineering, manuscript writing, and venue selection. I bring an unconventional background (four years of military service, three startups, and a semester at HKUST) that shapes how I identify problems worth solving.

Education

University of Cambridge Oct 2026 – Present
MPhil in Machine Learning and Machine Intelligence Cambridge, United Kingdom

The Hong Kong University of Science and Technology (HKUST) Aug 2025 – Dec 2025
Exchange Semester, Dept. of Computer Science and Engineering Hong Kong SAR, China

- Specialisation courses: COMP4211 Machine Learning, COMP4471 Deep Learning in Computer Vision, COMP4901B Large Language Models, COMP4901Z Reinforcement Learning.
- Postgraduate course: COMP6411D Data Visualisation.

University of Southern Denmark (SDU) Sep 2023 – Jun 2026
Bachelor of Science (BSc), Software Engineering Sønderborg, Denmark

- Expected graduation: June 2026. Current GPA: $\approx$ 10.6/12 ($\approx$ 3.7/4.0 US GPA).
- Practical, project-centric curriculum with mandatory semester-long team projects on complex, data-intensive software systems in domains such as IoT and AI, often in collaboration with industry partners.

Selected Publications

2026-01 **P. M. Konrad**, T. Tanyel, and S. Ayvaz. Routing Subspaces: Auditing Evaluation-to-Deployment Mismatch in Fine-Tuned Language Models. Under review, *NeurIPS 2026*.
TL;DR. Path-patching localises an eval-vs-deployment signal to a narrow mid-depth attention window. Clamping a low-dimensional subspace closes the gap in 11 of 12 cells.

2026-01 **P. M. Konrad**, T. Tanyel, and S. Ayvaz. Acceptance Cards: A Four-Diagnostic Standard for Safe Fine-Tuning Defense Claims. Under review, *NeurIPS 2026*.
TL;DR. A four-gate evidential standard for safe-fine-tuning defense claims. SafeLoRA fails the full-card pass under matched-action audit.

2026-01 **P. M. Konrad**, T. Tanyel, and S. Ayvaz. Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout. Under review, *ICML 2026 Workshop on Mechanistic Interpretability*.
TL;DR. Moral framing is linearly decodable in pretrained models but only causally routed to judgment after instruction tuning.

2026-01 **P. M. Konrad** and S. Ayvaz. When Does Chain-of-Thought Improve Safety? Evidence from 18 Models Across 5 Families. Under review, *COLM 2026*.

TL;DR. CoT does not automatically improve safety calibration. Training objective (GRPO) matters more than deliberation itself.

2026-01 **P. M. Konrad**, T. L. Adam, A. C. H. Merrild, R. de Rosa, R. Terrenzi, T. Tanyel, and S. Ayvaz. The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime. Under review, *NeurIPS 2026*.

TL;DR. Position paper: mechanistic interpretability alone should not gate deployment. We propose a calibrated verification regime with a Verification Coverage metric.

Full Publication List

1. **P. M. Konrad**, T. Tanyel, and S. Ayvaz (2026). *Routing Subspaces: Auditing Evaluation-to-Deployment Mismatch in Fine-Tuned Language Models*. Under review, *NeurIPS 2026*.
2. **P. M. Konrad**, T. Tanyel, and S. Ayvaz (2026). *Acceptance Cards: A Four-Diagnostic Standard for Safe Fine-Tuning Defense Claims*. Under review, *NeurIPS 2026*.
3. **P. M. Konrad**, T. L. Adam, A. C. H. Merrild, R. de Rosa, R. Terrenzi, T. Tanyel, and S. Ayvaz (2026). *The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime*. Under review, *NeurIPS 2026*.
4. **P. M. Konrad**, Y. Du, and S. Ayvaz (2026). *Grounded Auditing as an Evaluation Policy: A Matched-Action Protocol and Stress Test*. Under review, *NeurIPS 2026*.
5. **P. M. Konrad** and S. Ayvaz (2026). *When Does Chain-of-Thought Improve Safety? Evidence from 18 Models Across 5 Families*. Under review, *COLM 2026*.
6. **P. M. Konrad**, Y. Du, T. Tanyel, and S. Ayvaz (2026). *How Much of a Probe Direction Lives in the Input Embedding? An Audit of Four Published Claims*. Under review, *EMNLP 2026*.
7. **P. M. Konrad**, Y. Du, T. Tanyel, and S. Ayvaz (2026). *Abstention and Refusal Are Separable Directions in the Residual Stream*. Under review, *EMNLP 2026*.
8. **P. M. Konrad**, T. Tanyel, and S. Ayvaz (2026). *Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout*. Under review, *ICML 2026 Workshop on Mechanistic Interpretability*.
9. **P. M. Konrad**, T. Tanyel, and S. Ayvaz (2026). *Counterfactual Self-Reports Are Not Well-Posed: A Mechanism-Binding Test for LLM Introspection*. Under review, *PhilML Workshop, ICML 2026*.
10. **P. M. Konrad**, T. Tanyel, and S. Ayvaz (2026). *A Path Already Walked: On Inheriting Network-Neuroscience Tools for Mechanistic Interpretability*. Under review, *ICML 2026 Workshop on Mechanistic Interpretability*.
11. N.-T. Thinh, Y. Du, **P. M. Konrad**, and A. Narechania (2026). *Fact-check Your Information (FYI): A Design Probe to Understand How People Actually Fact-check Data-Driven Articles*. Under review, *IEEE VIS 2026*.
12. **P. M. Konrad**, T. L. Adam, R. Terrenzi, and S. Ayvaz (2025). *Architecture Without Architects: How AI Coding Agents Shape Software Architecture*. Accepted, SAGAI Workshop, IEEE ICSA 2026.
13. T. L. Adam, **P. M. Konrad**, R. Terrenzi, F. G. Lukas, R. Yilmaz, K. Sierszecki, and S. Ayvaz (2025). *CAKE: Cloud Architecture Knowledge Evaluation of Large Language Models*. Accepted, KDA-AI Workshop, IEEE ICSA 2026.
14. R. Terrenzi, **P. M. Konrad**, T. L. Adam, and S. Ayvaz (2025). *A Reference Architecture for Agentic Hybrid Retrieval in Dataset Search*. Accepted, SAML Workshop, IEEE ICSA 2026.
15. **P. M. Konrad**, A.-A. Popa, Y. Sabzehmeidani, L. Zhong, M. Tripathy, A. Constantinescu, E. A. Liehn, and S. Ayvaz (2025). *Challenges in Deep Learning-Based Small Organ Segmentation: A Benchmarking Perspective for Medical Research with Limited Datasets*. Under revision, *Biomedical Signal Processing and Control*; arXiv:2509.05892.
16. **P. M. Konrad**, Y. Sabzehmeidani, A. Popa, and S. Ayvaz (2025). *Machine Learning in Gastrointestinal Tract Imaging: A Comprehensive Review of Techniques and Applications*. Under review, *Journal of Imaging Informatics in Medicine*.
17. **P. M. Konrad**, C. H. Kunstmann-Olsen, J. Fiutowski, and S. Ayvaz (2025). *Non-Destructive Prediction of Fruit Ripeness and Firmness Using Hyperspectral Imaging and Lightweight Machine*

Learning Models. Under review, *Computers and Electronics in Agriculture.*

18. **P. M. Konrad**, T. Tanyel, and S. Ayvaz (2025). *Beyond Major Floods: Deep Learning for Detecting Shallow Water Inundation in Agricultural Areas.* In *KES 2025*, Procedia Computer Science, 270, 301–310. [open access]
19. **P. M. Konrad** (2026). *When AI Coding Studies Disagree: A Practical Guide to the Evaluation Lens.* Under review.
20. **P. M. Konrad** and T. L. Adam (2026). *The Trader's Trinity: Forecasting Models, RL Agents, and LLM Judges for Day-Ahead Markets.* BSc thesis, University of Southern Denmark. In progress.

Research Experience

DataVISards, Hong Kong University of Science and Technology (HKUST) Jan 2026 – Present
Research Collaborator Hong Kong SAR, China

- Collaborating with the DataVISards research group on machine learning and data visualisation projects.

Data and Intelligence Lab, SDU 🌐 Sep 2024 – Present
Research Collaborator Jan 2026 – Present Sønderborg, Denmark

- Independently initiate and drive machine learning research projects: problem formulation, experimental design, pipeline development, manuscript writing, and venue selection.
- Co-author papers with Assoc. Prof. Serkan Ayvaz and lab members across remote sensing, medical imaging, NLP, and software systems.
- Lead the full research lifecycle end-to-end without external task assignment.

Research Assistant Sep 2024 – Dec 2025

- Built end-to-end ML pipelines and contributed to study design, literature reviews, model development, and manuscript preparation.
- Contributed to grant proposal preparation and multimodal dataset management (remote sensing, hyperspectral, medical imaging).
- Co-authored the KES-2025 paper on deep learning for shallow water inundation detection.

Academic Service

Teaching Assistant, Artificial Intelligence (SDU) 🌐 Jan 2026 – Jun 2026

- Teaching Assistant for the BSc-level course "Artificial Intelligence," supporting exercise sessions and student supervision.
- Built an interactive AI course covering AI fundamentals with step-by-step animations, used as supplementary material for the course.
- Responsibilities include designing hands-on lab assignments, providing student support, and contributing to curriculum development.

Educational Committee, Software Engineering Programme, SDU Jun 2025 – Present
Student Member Sønderborg, Denmark

- Represent BSc Software Engineering students in curriculum and quality-assurance discussions, providing feedback on course content, assessment, and study environment.

Professional Experience

SaturoLabs 🌐 Jan 2026 – Present
Founder Denmark

- Founded and currently leading SaturoLabs, building products at the intersection of AI and software engineering.
- Live products: claudeboyz.com, getproofz.com, dreambear.app.

Tutora ApS 🌐 Jun 2024 – Nov 2024
CTO and Co-Founder Sønderborg, Denmark

- Led the end-to-end development of the company's websites and core web application.
- Implemented the Shape Up product development framework to streamline technical execution.

Yeager GmbH Nov 2022 – Jun 2024
Co-CEO and Co-Founder Remote

- Co-founded the company and led development of *stabil.ai*, an AI-powered mobile app for personalized powerlifting training.
- Engineered algorithms to personalize training plans using MRV/MEV and dynamic real-time feedback.
- Oversaw the full product lifecycle, from UX/UI concept and design to full-stack implementation.

Transition Year: Independent Study & Prototyping Oct 2021 – Oct 2022
Independent Germany/Denmark

- Completed intensive coding bootcamps and self-directed study in software engineering to transition from military service into the technology field.
- Began prototyping early concepts for a training-app product, which later evolved into the product developed at Yeager GmbH.

Bundeswehr (German Armed Forces) Oct 2017 – Sep 2021
Staff Duty Soldier Glücksburg, Germany

- Led a small HR team responsible for the administration of over 600 soldiers.
- Streamlined administrative processes and document workflows in a high-stakes naval command environment.

Telecommunications Company Jul 2016 – Sep 2017
Customer Service Consultant Germany

- Provided front-line customer support for mobile and broadband services.

Skills

Programming Languages: Python, JavaScript, SQL
ML Libraries: PyTorch, Hugging Face, TransformerLens, Scikit-learn, Pandas, NumPy
Data Visualization: Matplotlib, Seaborn
ML Engineering: Data & Feature Engineering, Hyperparameter Tuning (Optuna, Weights & Biases Sweeps), Experiment Tracking (Weights & Biases)
Web Development: React, React Native, Node.js, Express, HTML/CSS, Streamlit
Tools & Platforms: Git, Docker, Google Cloud Platform, Vercel
Methodologies: Scrum, Shape Up

Languages

English (Professional working)
German (Native or bilingual)
Thai (Native or bilingual)
Danish (Elementary)

Honors and Awards

William Demant Fonden Grant May 2026
Received a grant from the William Demant Foundation supporting graduate studies.

Top-10 Placement, Danish National Championship in AI (DM i AI) 🌐 2025
National AI competition, individual competitor

- Secured a top-10 placement while competing solo against more than 50 teams in Denmark's largest AI competition, organized by the Danish Society for Artificial Intelligence.

Top-10 Placement, Danish National Championship in AI (DM i AI) 2024
National AI competition, individual competitor

- Achieved top-tier placement in a national AI competition for students, organized by the Danish Society for Artificial Intelligence.

1st Place, SDU Case Competition, SDU Sønderborg 2023
University case competition on sustainability

- Awarded 1st place in a 48-hour competition focused on sustainability with real-world cases from companies such as Danfoss and Linak.

Formal Recognition for Exemplary Service, Bundeswehr (German Armed Forces) 2021
Military commendation

- Formally commended for setting a benchmark in dedication and professionalism for enlisted personnel.

Performance Bonus for Outstanding Achievement, Bundeswehr (German Armed Forces)
2021
Military achievement bonus

- Awarded a financial bonus for sustained, excellent performance and independent handling of complex tasks.

Certifications

Venture Capital Explorer Programme, Accelerace & BII 2026
Intensive 4-day program covering VC investment process, founder sourcing, startup assessment, venture risk evaluation, investment committee operations, exits, and case-study simulations.

Machine Learning, Stanford Online 2024
Foundational C# with Microsoft 2024
Google UX Design Professional, Google 2023
React Native Course 2023
Full-Stack Engineer Career Path, Codecademy 2023
Certified Specialist for Real Estate Loan Brokerage (IHK) 2022